

Satyabrata Das

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Portfolio: satyabratadas.github.io | Github: [Satyabratadas](https://github.com/Satyabratadas) | LinkedIn: [satyabrata-lm10](https://www.linkedin.com/in/satyabrata-lm10)

Gainesville, FL, USA

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Education

University of Florida, Gainesville, FL

Aug 2025 – May 2027

M.S. in Artificial Intelligence Systems

- Relevant Coursework: AI Safety & Robustness, AI Systems, Machine Learning for AI, Image Processing & Computer Vision, Applied Deep Learning

Guru Nanak Institute of Technology, India

Aug 2018 – Jun 2022

B.Tech in Computer Science | CGPA: 8.64 / 10

Skills

Programming & Core CS: Python, Swift, C++, SQL, Data Structures, Algorithms, OOP, Git

Machine Learning & NLP: PyTorch, scikit-learn, Transformers, NLP, NumPy, Pandas

Computer Vision: Filtering (spatial/frequency), edge detection, morphology, segmentation, object recognition
Systems & iOS: FastAPI, REST APIs, Microservices, Docker, Prometheus/Grafana, SwiftUI, Xcode, Core Data

Experience

ARC Document Solutions, India

Jan 2022 – Sep 2024

Software Engineer

- Built and scaled real-time collaboration systems (WebSocket-based changesets, QR scanning modules, REST APIs), reducing sync latency by **25%** for over **1,000+** daily active users.
- Developed an iOS-based Wi-Fi security analyzer using Swift and NetworkExtension to detect WPA/WPA2/WPA3 vulnerabilities, identifying risks across **20+** enterprise devices.
- Enhanced ARC Print's checkout reliability by integrating Paytm, Cashfree, Razorpay, Braintree, and Elavon payment gateways along with Google reCAPTCHA, reducing failed transactions by **15%**.
- Designed and shipped a high-performance SwiftUI filtering system, improving product discovery and boosting order conversion by **10%** for more than **5K** monthly users.
- Collaborated cross-functionally with QA, backend, DevOps, and design teams to deliver secure, scalable iOS applications, contributing to a **30%** drop in customer-reported defects.

SCI-BI Software Pvt Ltd, India

Jul 2021 – Dec 2021

Software Trainee

- Engineered high-volume ETL pipelines (SSIS, SSMS, Python) processing millions of records daily with **99%+** accuracy.
- Automated data workflows using Python, cutting manual effort by **40%** and saving **20+ hours/week**.
- Optimized SQL queries and ETL transformations, improving execution speed by **30%** and enabling faster BI insights.

Certifications & Achievements

- **Hacklytics 2026 — Participant**
- **NASA EMERGE Data Hackathon — Winner**
- **Supervised Machine Learning: Regression and Classification — DeepLearning.AI**
- **Python for Data Science, AI & Development — IBM**

[See Full list at Portfolio]

Selected Projects

SummarIQ: Math-Aware Research Paper Summarizer (*NLP, LLMs, LaTeX Parsing*)

- Engineered a **math-aware scientific summarization system** integrating LaTeX parsing, equation ranking, and transformer-based generation (T5/BART), producing section-wise summaries and equation explanations.
- Implemented a multi-service FastAPI backend with Docker, achieving **<1.4s average latency** and robust Prometheus/Grafana instrumentation (latency, GPU utilization, drift).
- Devised a custom LaTeX extraction pipeline that processed **200+ academic papers**, identifying equations, figures, and tables with **< 92% accuracy**.

JuggleIQ: CV-Based Soccer Performance Analytics System (*OpenCV, YOLO, MediaPipe*)

- Constructed a computer vision pipeline integrating YOLO object detection, MediaPipe pose estimation, and Kalman filtering to process **30–60** second juggling videos, generating **1,000+** frame-level inferences per session.
- Formulated a GPU-accelerated FastAPI backend that returns structured analytics across 6+ performance metrics, including skill score, **L/R** touch distribution, rhythm intervals, peak height, lateral drift, and knee-angle feedback.
- Delivered algorithmic touch classification and scoring models, providing quantitative performance insights and coaching recommendations visualized through an interactive React dashboard.